

Work through the following exercises while the TA circulates. When you have completed the exercises, spend the rest of the laboratory working on the homework assignment.

Warm up.

Recall that the definition of the derivative is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Explain using a picture why this definition gives the slope of the tangent line to $f(x)$ at the point $(x, f(x))$.

Exercise 1.

Using the above definition of the derivative, compute the derivative of the following functions.

1. $f(x) = \sqrt{x}$.
2. $f(x) = \frac{1}{x}$.

Exercise 2.

Recall that a line can be given as a function $L(x) = mx + b$, where m is the slope of the line and b is given by the point where the line intersects the y -axis. Using your answer from the previous exercise, find the equation of the line tangent to the following functions at the following points.

1. $f(x) = \sqrt{x}$ at $(1, 1)$.
2. $f(x) = \frac{1}{x}$ at $(1, 1)$.